PURDUE
UNIVERSITY

School of Materials Engineering

MSE 250: Physical Properties in Engineering Systems
Spring 2026

Course Information

MSE 25000 Physical Properties in Engineering Systems
CRN 17618
Meeting time: MWF, 11:30 am – 12:20 pm, ARMS 1109
Instructional modality: Face-to-Face
Course credit hours: 3
Course online page: Brightspace
Prerequisites: Prerequisites: PHYS 17200, Corequisites: MSE 23000, MA 26500 (or MA 26200)
Course Status: MSE 25000 is a required course

Information About the Instructor(s)

See Purdue Class Schedule for spring 2026

Catalog Description

Class connects math, science and engineering practice and applications. Presents foundational aspects of engineering problem solving, use of computer math tools for engineering problem solving, basic engineering statics, dynamics and mechanics, introduction to stress tensors, group problem solving approaches, and introductory aspects of design and materials selection.

Topics Covered:

Stress and Strain in 1d, 2d, and 3d, stress and strain tensors, tensor differentiation, rotation of tensors.
Statics including: free body diagrams, bending stresses, beams with simple, cantilevered and distributed loadings, shear and moment diagrams for beams, bending and shear stresses in beams as a function of cross-section, position in the beam, and position through the beam thickness. Brittle and ductile failure, torque, torsional strain and displacement, elastic modulus and stress-strain behavior, thin-walled pressure vessels.

Learning Outcomes (Goals of the course and student outcomes)

<u>All Students</u>

A. Construct free body diagrams for static loading that include both forces and moments.

B. Be able to mathematically write and solve simple mathematical relationships that pertain to a physical situation including:

- Be able to write force and moment vectors and manipulate them.
- Be able to write force reactions in x, y, and z directions and solve them simultaneously.
- Perform 2-D force rotations to a new coordinate axis.
- Calculate bending, torsional, transverse shear, and tension stresses and deflections for a variety of loading conditions.

- Determine if a material, with a known yield strength, will yield given the principal stresses.

C. Be able to interpret the effects of loading on a beam including:
- Draw the shear and moment diagrams for a loaded beam.
- Be able to draw the shear and moment diagrams for 3 and 4 point flexure and discuss differences during mechanical testing.

Most Students

A. Be able to design a component that is part of a larger system including:
- Be able evaluate which type of stress (bearing, shear, bending, tension) will cause failure for a member subject to stresses from forces and moments.
- Design the cross-section of a beam to meet certain design criteria.

B. Be able to analyze a complex loading state including:
- Being able to identify which stresses act on an element.
- Be able to construct a 3-dimensional element and correctly place the stresses on it.
- Be able to write the stress tensor for a complex loading state.
- Be able to write the rotation matrix for a stress tensor to solve for principal stresses.

Relation of Course to Student Outcomes

(MSE-1, ABET-1) an ability to identify, formulate, and solve complex materials engineering problems by applying principles of engineering, science, and mathematics.

Teaching Philosophy

NSE250 is an important foundational course. It is my goal to ensure that the students grasp all concept of mechanics of materials introduced in this course in order to prepare them for more advanced topics in mechanical behavior of materials and the use of materials in design. The student performance in future topics such as deformation, creep and fracture of structural metals, ceramics and polymers will depend on the basic mechanics concepts explained in this course. With this in mind, I will always try to highlight those connections as we discuss various parts of the course. Mechanics of materials requires solving practice problems frequently. I have designed my lectures to explain the concepts through problem solving. Homework assignments are designed to add further hands-on experience. I highly value the interaction with me during class time and office hours.

How to Succeed in this Course

To ensure success in both learning and in getting high grades in this course, please be willing to speak up when you need help with the course material, manage your time wisely, and do your reading and homework assignments before last minute.

Learning Resources, Technology, & Texts

Textbook: Russell C. Hibbeler, Statics and Mechanics of Materials, 6th Edition, 2024. (5th Edition is also okay.)

Class Notes: The instructor notes will be made available to students via Brightspace.

Copyright

Do not share my course notes, homework assignments, solutions, or exams with anyone or any entity outside of this course. This includes posting to websites that are available online. Material of this course that I distribute is copyrighted (by me or by the authors of the book) and may not be reproduced without permission. See the *University Policies and Statements* section of Brightspace for guidance on Use of Copyrighted Materials. Effective learning environments provide opportunities for students to reflect, explore new ideas, post opinions openly, and have the freedom to change those opinions over time. Students and instructors are the authors of the works they create in the learning environment. As authors, they own the copyright in their works subject only to the university's right to use those works for educational purposes. Students may not copy, reproduce, or post to any other outlet (e.g., YouTube, Facebook, or other open media sources or websites) any work in which they are not the sole or joint author or have not obtained the permission of the author(s).

Instructor's Office Hours

MWF 2:00 pm – 3:00 pm, 2215 ARMS
Online Zoom link: https://purdue-edu.zoom.us/j/5211402390. Will open Zoom upon request.

Instructor's Email Availability and Policies

I will be available via email daily, and I will try to respond within two hours or less, between 8 am and 8 pm. When emailing me, please use MSE 250: [Issue] in the subject line of the email, e.g., MSE 250: Homework #3 question.

Assignments and Points

Assignments	Date/Time	Weight
Exam 1	Friday 02/20	10
Exam 2	Wednesday 03/25	25
Exam 3	Friday 04/24	25
Homework	Throughout semester (see next table)	40%
Final Exam	None	
		Total: 100

- Exams are not cumulative.
- There will be no make-up exams except in extreme circumstances. The student must inform the instructor in advance by email or in person that he/she needs to take an exam not at the stated dates.
- About 6 homework assignments will be given. In some assignment, students are expected to use MATLAB or equivalent software to obtain results.
- Unless stated otherwise, homework assignments are automatically due a week from the day they are assigned.
- Homework submitted after due date will be considered late and will receive less or no credit. Exceptions can be made in certain circumstances with proper documentation and instructor approval.

Detailed instructions regarding turning in the homework/exam and seeing your graded work and grades will be given with homework #1. All submitted homework assignments must:

- have a cover page,
- have the problems in the correct order,
- be written legibly on clean paper,
- have sufficient work to justify solution, and
- have the solution clearly marked

Students are encouraged to study and discuss homework problems together to strengthen collaborative learning. However, copying others' work will justify grade deductions.

Grading Scale

A (90^+ to 100); A^-(85^+ to 90)
B^+ (80^+ to 85); B (75^+ to 80); B^- (70^+ to 75)
C^+ (65^+ to 70); C (60^+ to 65); C^- (55^+ to 60)
D (50^+ to 55); F <49%

Incompletes

A grade of incomplete (I) will be given only in unusual circumstances. To receive an "I" grade, a written request must be submitted prior to December 1, and approved by the instructor. The request must briefly describe the circumstances, along with a proposed timeline for completing the course work. Submitting a request does not ensure that an incomplete grade will be granted. If granted, you will be required to fill out and sign an "Incomplete Contract" form that will be turned in with the course grades. Requests made after the course is completed will not be considered for an incomplete grade.

Course Schedule

The schedule below will be followed to the instructor's best. However, minor changes in delivery and homework assignment dates are to be expected. Exams 1, 2 and 3 dates are fixed and will not change.

Week	Date (Monday)	Topics for the week (Minor changes possible)	Remarks
1	01/12	Syllabus discussion, Introduction, Units, Review of vectors	
2	01/19	MLK Day (01/19) Force vectors	HW-1
3	01/26	Force system resultants	HW-2
4	02/02	Equivalent force systems/Moments and couples Moments in 3D	
5	02/09	Equilibrium of a rigid body 2D Problems Equations of equilibrium	HW-3
6	02/16	Free body diagrams (FBD) Exam 1, Friday 02/20 Review problem set will be distributed a few days before exam	**In-class Exam 1** **Friday 02/20**
7	02/23	Geometric principles and distributed loading	HW-4
8	03/02	Stress and strain	HW-5
9	03/09	Mechanical properties of materials	
10	03/16	Spring Break	
11	03/23	Axial loading Exam 2, Wednesday 03/25 Review problem set will be distributed a few days before exam	HW-6 **In-class Exam 2** **Wednesday 03/25**
12	03/30	Axial loading	

		Torsion	
13	04/06	Torsion Bending	HW-7
14	04/13	Bending Combined loading	
15	04/20	Combined loading Exam 3, Friday 04/24 Review problem set will be distributed a few days before exam	**In-class Exam 3** **Friday 04/24** HW-8
16	04/27	Yield criteria and related design issues Week before finals week	
17	05/04	Finals Week This class has no final	

Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform me of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact me as soon as possible by Purdue email or phone. For absences that do not fall under excused absence regulations (see below), this course follows the following procedures:

Do not come to class if you are feeling ill but DO email me at aelazab@purdue.edu, with the subject line: MSE 508 absence. I do not need details about your symptoms. Just let me know you are feeling ill and cannot come to class. If it is an emergency, please follow the University regulations on medical care (see below).

Unless it falls under the University excused absence regulations (see below), any work due should be submitted on time via our course Brightspace.

If that day's class involves assessed work such as a test or presentation, you and I will plan if and how you can make up the work, following the assignment guidelines. This plan must be done before the next class period, so again, email me immediately when you know that you will miss the class.

The most important consideration in any absence is how it will affect your achievement of the assignment objectives and the course learning outcomes.

For cases that fall under **excused absence regulations**, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or certain types of medical care. The processes are detailed, so plan ahead.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-

8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on the Academic Resources table on your Brightspace homepage.

You may use AI to study but the use of AI in solving homework assignments or during exams is not allowed.

The Purdue Honor Pledge "As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue"

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in the Academic Resources table on your Brightspace homepage.

Accessibility

Disability-related accommodations

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: https://www.purdue.edu/drc/students/course-accommodation-letter.php Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodations.

Accommodated testing

Students with disabilities whose DRC Course Accommodation Letter (CAL) includes test accommodations must first release their CAL to me and then schedule to take their exams with Purdue Testing Services at https://www.purdue.edu/studentsuccess/testing-services/accommodated-testing/student.php. You must schedule at least four days (96 hours) before the exam date listed on the syllabus.

Mental Health Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.
If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can

be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

See Purdue's Information on Emergency Preparation and Planning. This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the *Emergency Preparedness for Classrooms document*.

The first day of class, I will review the **Emergency Preparedness plan for our specific classroom**. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.

Disclaimer

This syllabus may be subject to change. Should any changes in the syllabus be made, they will first be discussed in class and then an updated version of the syllabus will be uploaded to Brightspace which will highlight the changes proposed or made.